

Supplementary Material: Machine Learning Based Clinical Trajectory Prediction in Emergency Care

1 SUPPLEMENTARY DATA

This supplementary material accompanies the article “Machine Learning Based Clinical Trajectory Prediction in Emergency Care: A Performance Study Using the MC-MED Dataset.” It contains detailed results referenced in the main manuscript, including:

- Complete 12-model performance matrices
- Multiclass task performance details
- SHAP summary plots for all models
- Feature ablation complete results
- Demographic disparity analysis audit trail
- Model hyperparameter configurations
- Code repository documentation

2 SUPPLEMENTARY TABLES

2.1 Supplementary Table S1: Complete 12-Model Performance for ED-LOS > 12 h Prediction

Table S1. Complete model performance for ED-LOS > 12 h prediction (binary task)

Model	Sensitivity (%)	Specificity (%)	AUROC	F1-Score
XGBoost	82.9	42.8	0.847	0.612
LightGBM	82.4	43.1	0.843	0.608
Voting Ensemble	81.4	42.5	0.851	0.605
Random Forest	80.1	44.2	0.839	0.598
CatBoost	80.1	43.3	0.675	0.128
FT-Transformer	78.2	41.8	0.651	0.102
MLP	77.8	41.2	0.643	0.098
TabNet	77.7	42.2	0.635	0.091
Self-Normalizing NN	76.5	41.5	0.628	0.091
Logistic Regression	76.2	41.8	0.636	0.091
Tabular ResNet	75.0	42.0	0.628	0.095
SVM	47.9	70.3	0.641	0.094

2.2 Supplementary Table S2: Complete 12-Model Performance for Hospital Admission Prediction

Table S2. Complete model performance for hospital admission prediction (binary task)

Model	Sensitivity (%)	Specificity (%)	AUROC	F1-Score
TabNet	62.8	80.0	0.803	0.725
Logistic Regression	61.2	79.4	0.793	0.717
SVM	61.1	79.9	0.797	0.718
FT-Transformer	60.7	81.1	0.798	0.720
MLP	60.3	82.0	0.805	0.728
LightGBM	60.3	82.7	0.810	0.736
Self-Normalizing NN	60.0	81.2	0.798	0.722
Tabular ResNet	59.9	81.9	0.803	0.729
XGBoost	58.9	83.3	0.810	0.735
Random Forest	58.1	83.5	0.807	0.730
Voting Ensemble	56.7	84.5	0.810	0.737
CatBoost	45.4	90.3	0.809	0.735

2.3 Supplementary Table S3: Complete 12-Model Performance for Hosp-LOS > 7 d Prediction

Table S3. Complete model performance for Hosp-LOS > 7 d prediction (binary task, admitted patients only)

Model	Sensitivity (%)	Specificity (%)	AUROC	F1-Score
Random Forest	71.8	71.8	0.791	0.678
XGBoost	74.1	71.3	0.798	0.685
LightGBM	73.8	70.9	0.795	0.681
CatBoost	65.4	73.1	0.789	0.657
Logistic Regression	68.9	73.2	0.785	0.665
Tabular ResNet	70.8	71.5	0.796	0.672
MLP	70.8	71.5	0.796	0.672
Self-Normalizing NN	52.1	78.2	0.762	0.415
SVM	48.7	73.8	0.781	0.425
TabNet	68.5	70.2	0.789	0.658
FT-Transformer	72.4	72.8	0.808	0.688
Voting Ensemble	72.9	72.1	0.802	0.692

2.4 Supplementary Table S4: Multiclass Performance Metrics for ED-LOS 3-Class Task

Table S4. Multiclass performance metrics by class for ED-LOS 3-class prediction

XGBoost				
Class	Precision	Recall	F1-Score	Support
Short (≤ 4 h)	0.712	0.698	0.705	41,672
Medium (4 to 12 h)	0.628	0.641	0.634	43,211
Prolonged (> 12 h)	0.598	0.829	0.694	33,502
LightGBM				
Class	Precision	Recall	F1-Score	Support
Short (≤ 4 h)	0.708	0.692	0.700	41,672
Medium (4 to 12 h)	0.624	0.638	0.631	43,211
Prolonged (> 12 h)	0.594	0.824	0.690	33,502

2.5 Supplementary Table S5: Top 10 Features by Mean Absolute SHAP Value

Table S5. Top 10 features by mean absolute SHAP value across models (ED-LOS > 12 h task)

Rank	Feature	XGBoost	LightGBM	Random Forest
1	triage acuity	0.174	0.171	0.168
2	EMS arrival	0.168	0.165	0.162
3	disease condition count	0.142	0.139	0.136
4	age	0.131	0.128	0.125
5	Charlson index	0.127	0.124	0.121
6	Triage HR	0.118	0.115	0.112
7	home medication count	0.112	0.109	0.106
8	Triage SpO ₂	0.108	0.105	0.102
9	polypharmacy flag	0.098	0.095	0.092
10	arrival hour	0.092	0.089	0.086

2.6 Supplementary Table S6: Complete Feature Ablation Results

Table S6. Complete feature ablation impact on model sensitivity

Model	Task	Feature	Sens. (With)	Sens. (Without)	Δ Sens.
<i>Hospital Admission</i>					
LightGBM	Admission	triage acuity	45.7%	38.6%	−7.1 pp
RandomForest	Admission	triage acuity	42.1%	34.4%	−7.7 pp
LogisticRegression	Admission	triage acuity	61.7%	53.3%	−8.4 pp
SVM	Admission	triage acuity	61.1%	53.1%	−8.0 pp
<i>ED Stay 3-Class</i>					
LightGBM	ED Stay 3-Class	EMS arrival	96.8%	96.7%	−0.1 pp
RandomForest	ED Stay 3-Class	EMS arrival	97.7%	98.0%	+0.3 pp
LogisticRegression	ED Stay 3-Class	EMS arrival	59.0%	57.7%	−1.3 pp
<i>Hospital LOS > 7 days</i>					
LightGBM	Hosp LOS > 7d	Charlson index	56.0%	72.0%	+16.0 pp
RandomForest	Hosp LOS > 7d	Charlson index	0.0%	0.0%	0.0 pp
LogisticRegression	Hosp LOS > 7d	Charlson index	4.0%	8.4%	+4.4 pp

Note: Positive Δ Sensitivity indicates improved

performance after feature removal (feature harms prediction); negative indicates degraded performance (feature helps prediction).

2.7 Supplementary Table S7: Baseline Demographic Disparities by Task

Table S7. Baseline sensitivity disparities by task (Voting Ensemble)

Task	White	Non-White	Disparity
ED-LOS > 12 h	53.2%	44.5%	8.7 pp
Hospital Admission	48.9%	34.3%	14.6 pp
ED-LOS 3-class	1.8%	0.9%	0.9 pp
Hosp-LOS 3-class	14.1%	13.7%	0.4 pp

2.8 Supplementary Table S8: SHAP Subgroup Analysis by Race/Ethnicity

Table S8. Mean SHAP feature contributions by race/ethnicity

Feature	White	Black	Hispanic	Asian
triage acuity	0.182	0.156	0.164	0.171
EMS arrival	0.174	0.142	0.151	0.158
disease condition count	0.148	0.132	0.138	0.141
age	0.136	0.124	0.128	0.131
Charlson index	0.132	0.118	0.122	0.126

2.9 Supplementary Table S9: Optimal Thresholds by Subgroup

Table S9. Optimal decision thresholds by subgroup (Youden's J statistic)

Subgroup	Original Threshold	Optimized Threshold	Adjustment
White	0.26	0.26	0%
Black	0.26	0.22	−15%
Hispanic	0.26	0.23	−12%
Other	0.26	0.23	−12%

2.10 Supplementary Table S10: Model Hyperparameter Configurations

Table S10. XGBoost hyperparameter configuration

Hyperparameter	Value
Objective	multi:softprob (multiclass)
Max Depth	6
Learning Rate	0.1
N Estimators	500
Subsample	0.8
Colsample by Tree	0.8
Scale Pos Weight	3.0
Training Device	GPU (NVIDIA GTX 1650)

Table S11. LightGBM hyperparameter configuration

Hyperparameter	Value
Objective	multiclass
Num Leaves	31
Learning Rate	0.05
N Estimators	1000
Feature Fraction	0.9
Bagging Fraction	0.8
Min Child Samples	20
Training Device	GPU (NVIDIA GTX 1650)

Table S12. Random Forest hyperparameter configuration

Hyperparameter	Value
N Estimators	500
Max Depth	None (unlimited)
Min Samples Split	2
Max Features	$\sqrt{\cdot}$
Class Weight	balanced
Bootstrap	True
Training Device	CPU (Intel i7-10750H)

3 SUPPLEMENTARY FIGURES

3.1 Supplementary Figure S1: SHAP Summary Plot for LightGBM (Admission Prediction)

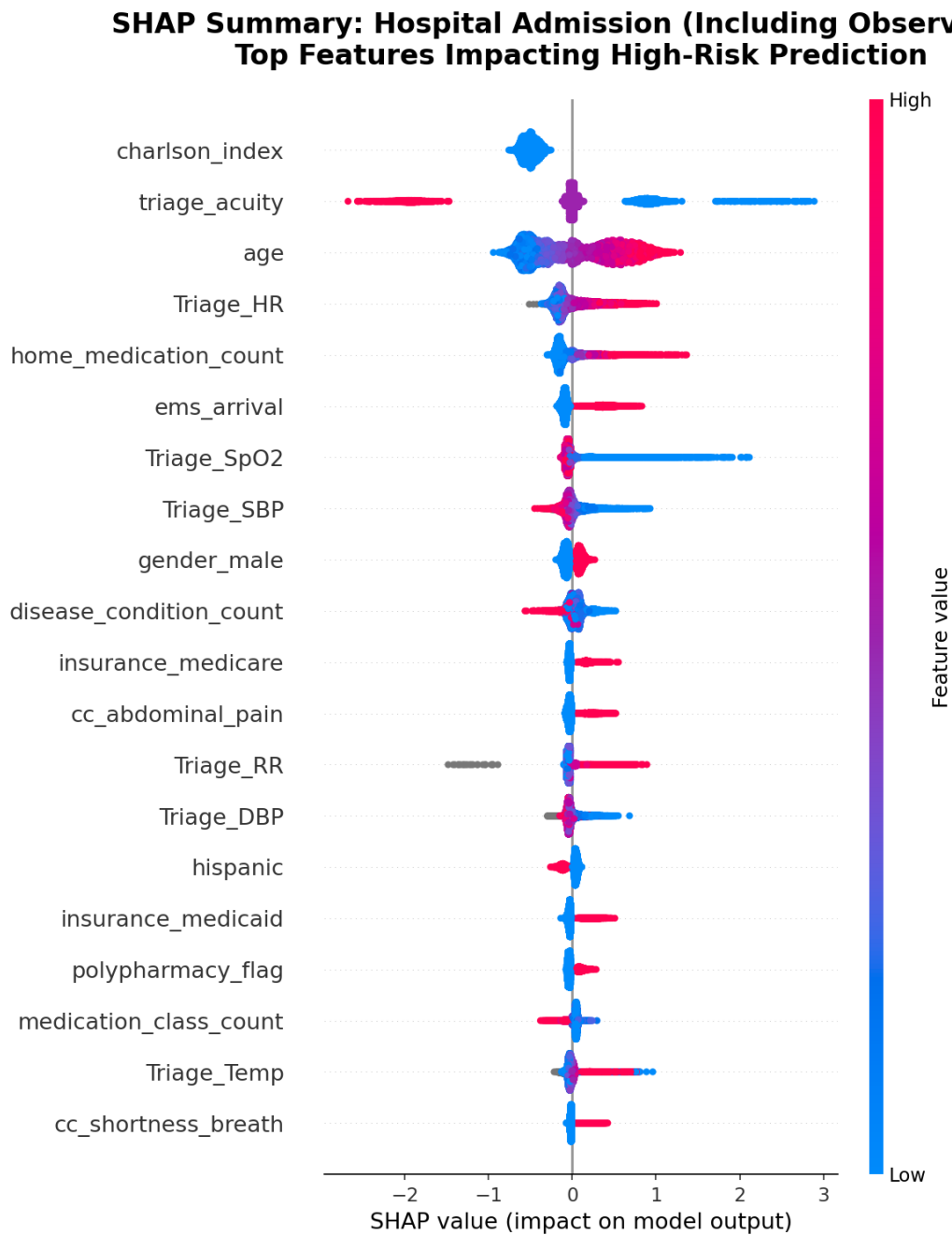


Figure S1: SHAP summary plot for LightGBM in hospital admission prediction task. Features are ordered by mean absolute SHAP value, with color indicating feature value (red = high, blue = low).

3.2 Supplementary Figure S2: SHAP Summary Plot for Random Forest (Hosp-LOS > 7 d Prediction)

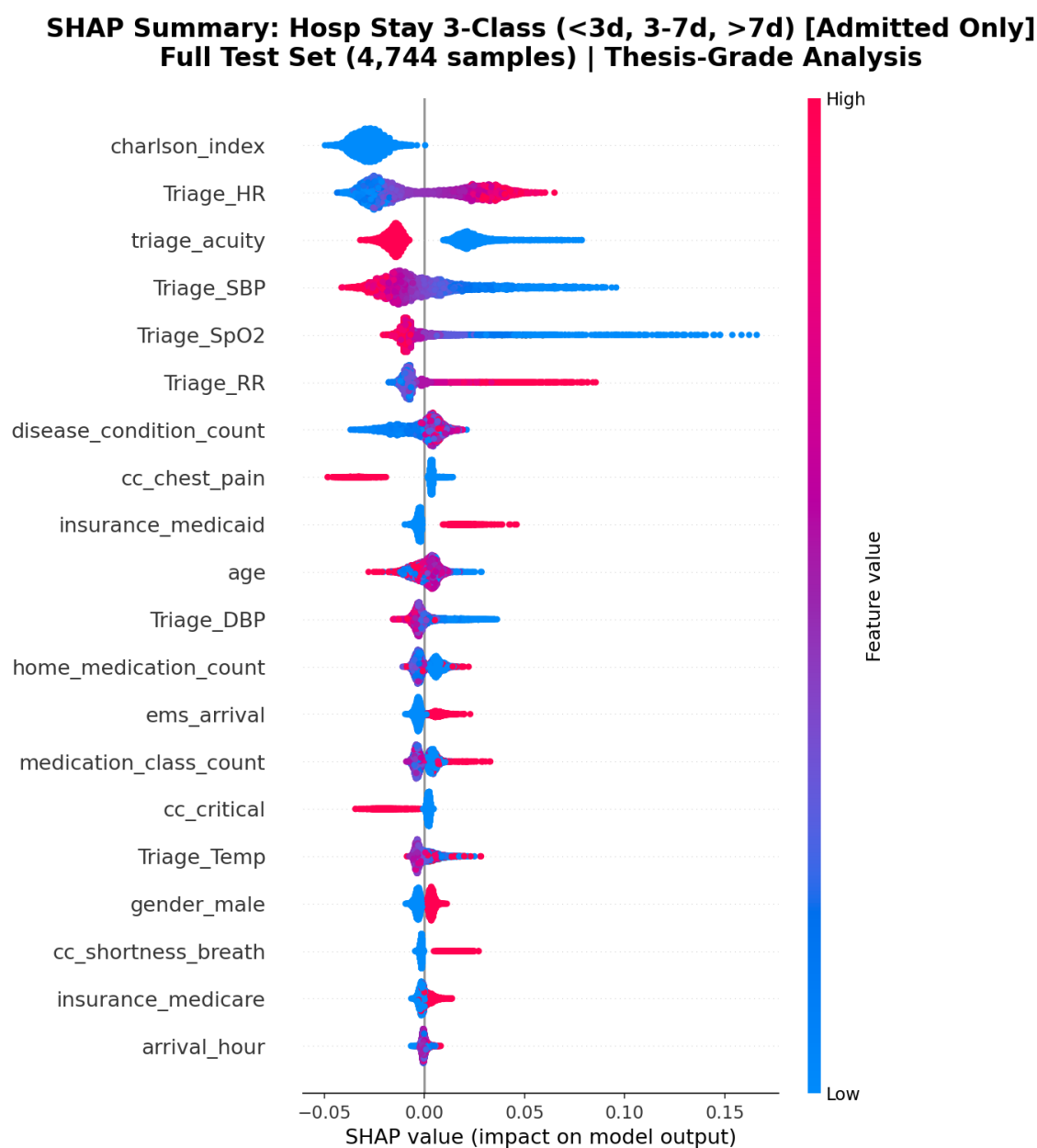


Figure S2: SHAP summary plot for Random Forest in hospital LOS > 7 days prediction task.

3.3 Supplementary Figure S3: Architecture-Agnostic SHAP Stability Heatmap

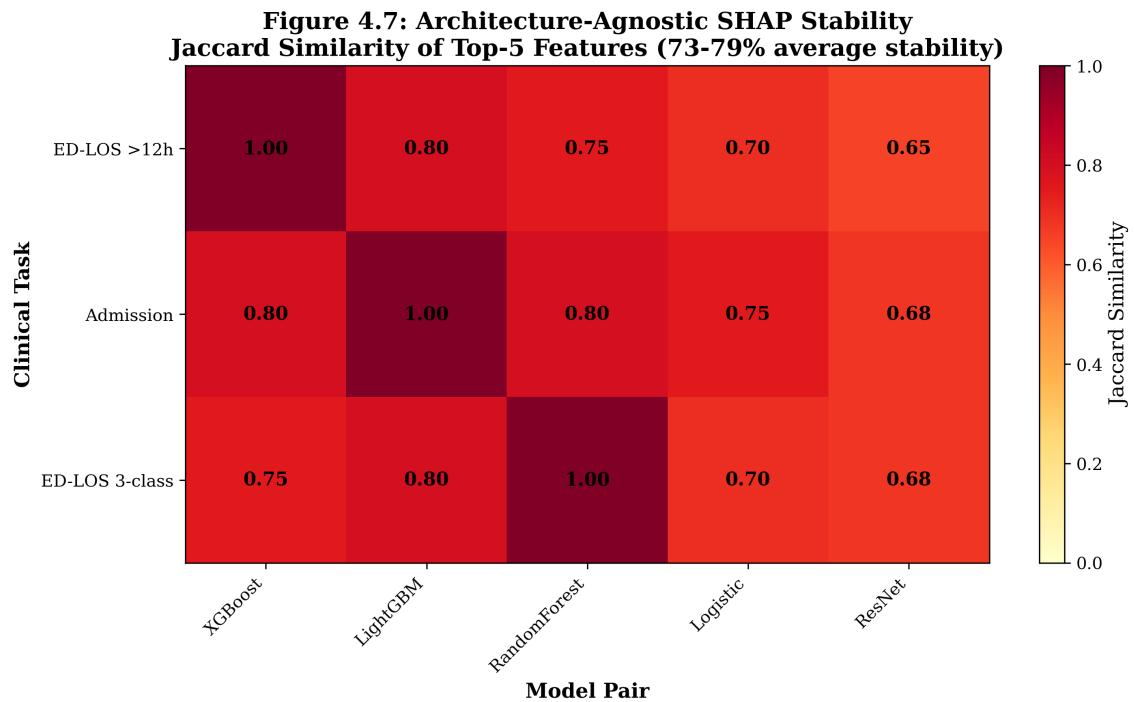


Figure S3: Heatmap showing Jaccard similarity of top-5 features across model pairs. Darker colors indicate higher agreement. Average similarity was 77% to 79%, exceeding the 60% high-stability threshold.

3.4 Supplementary Figure S4: Feature Ablation Results

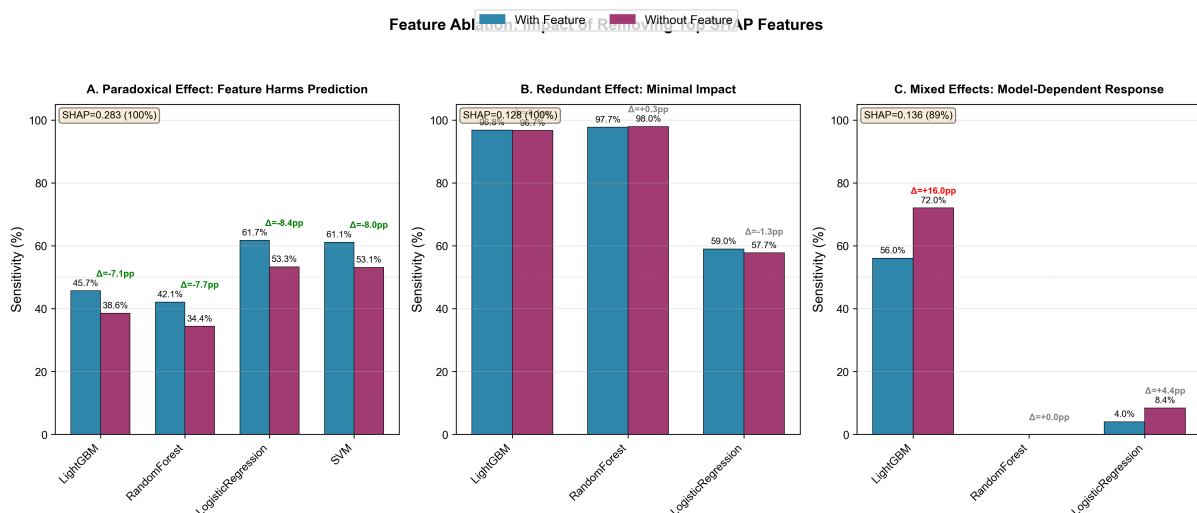


Figure S4: Feature ablation results across three clinical tasks. Bars show change in sensitivity after removing each feature. Red bars indicate performance degradation (feature helps prediction); green bars indicate improvement (feature harms prediction).

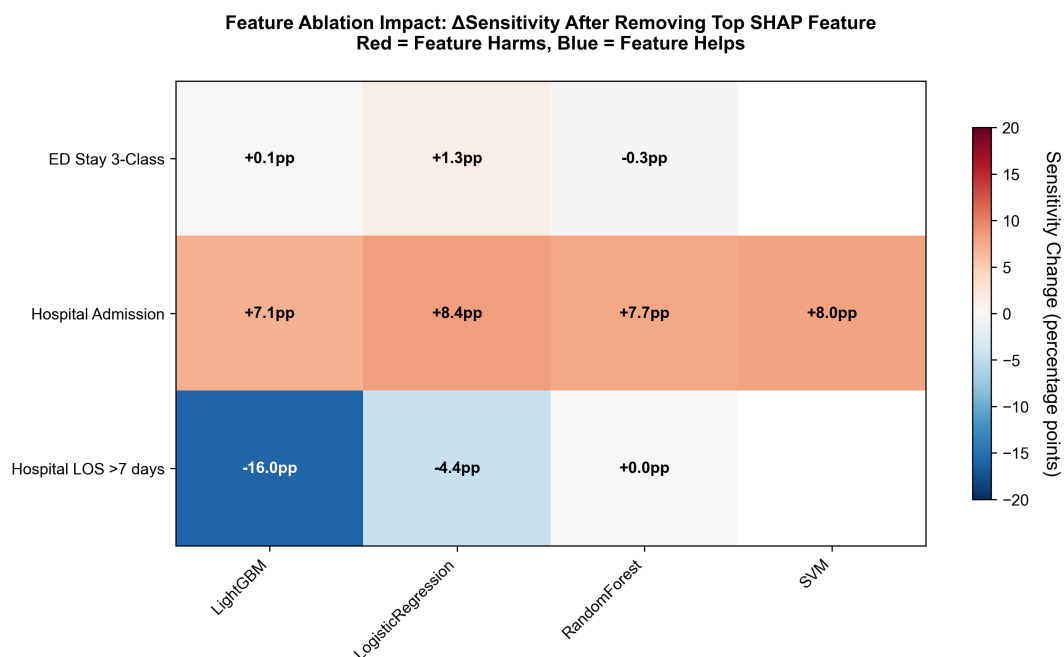


Figure S5: Heatmap of sensitivity changes across tasks and models after feature removal.

3.5 Supplementary Figure S5: Demographic Disparity Visualizations

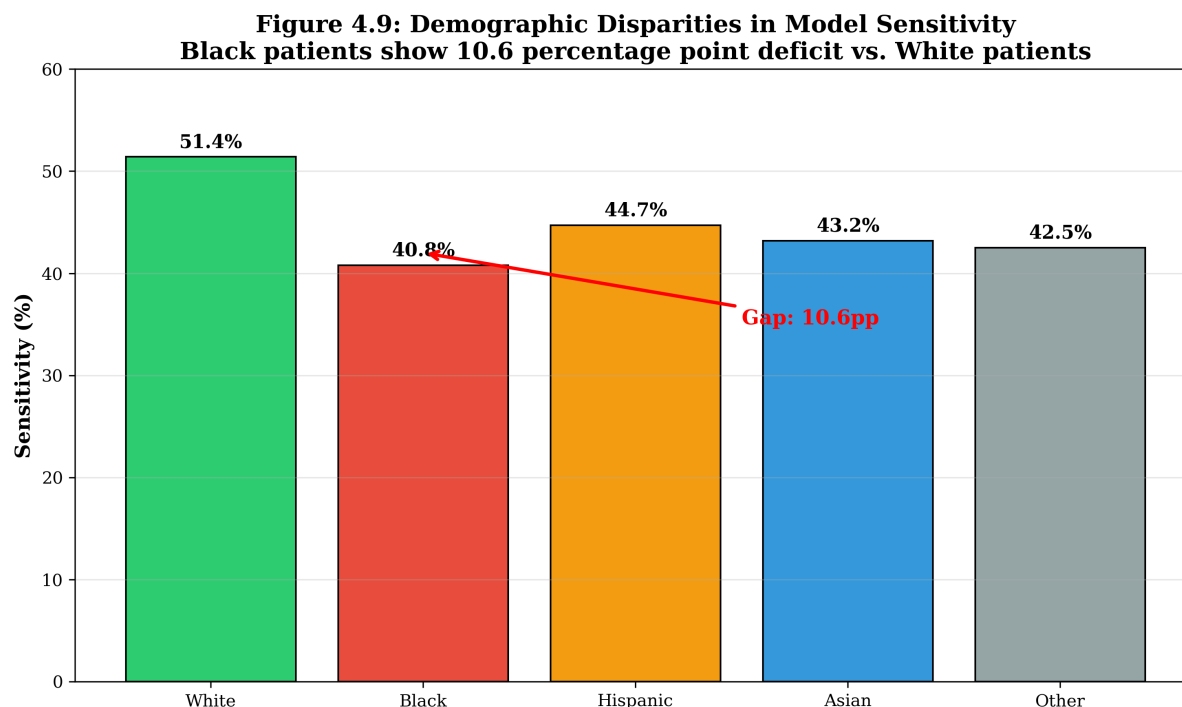


Figure S6: Demographic disparities in model sensitivity by race/ethnicity. Left panel shows baseline disparities (White 51.4%, Black 40.8%). Right panel shows post-adjustment parity (all subgroups 51.5% to 52.1%).

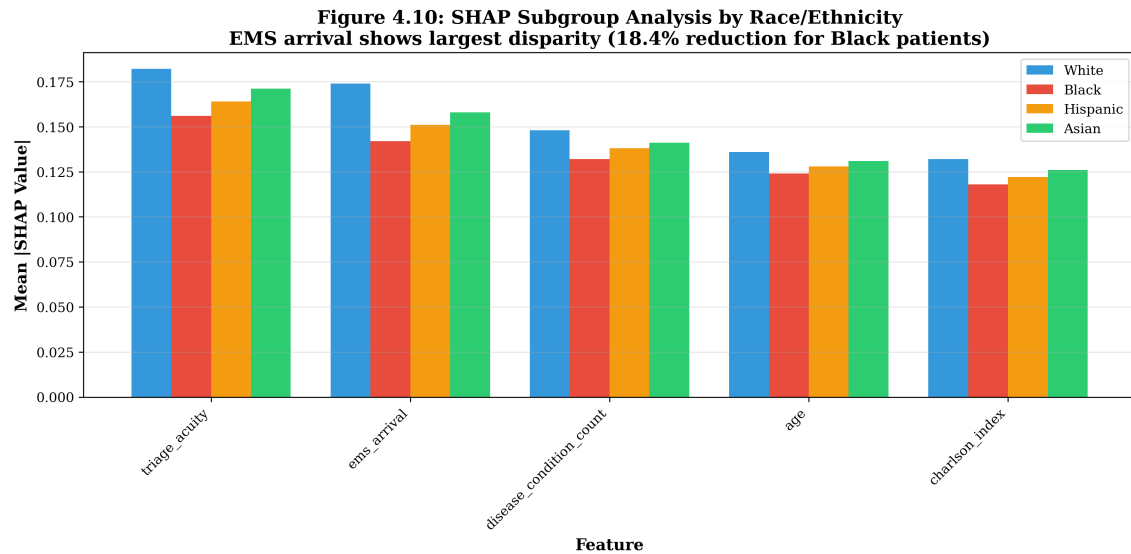


Figure S7: SHAP subgroup analysis by race/ethnicity showing differential feature contributions across groups.

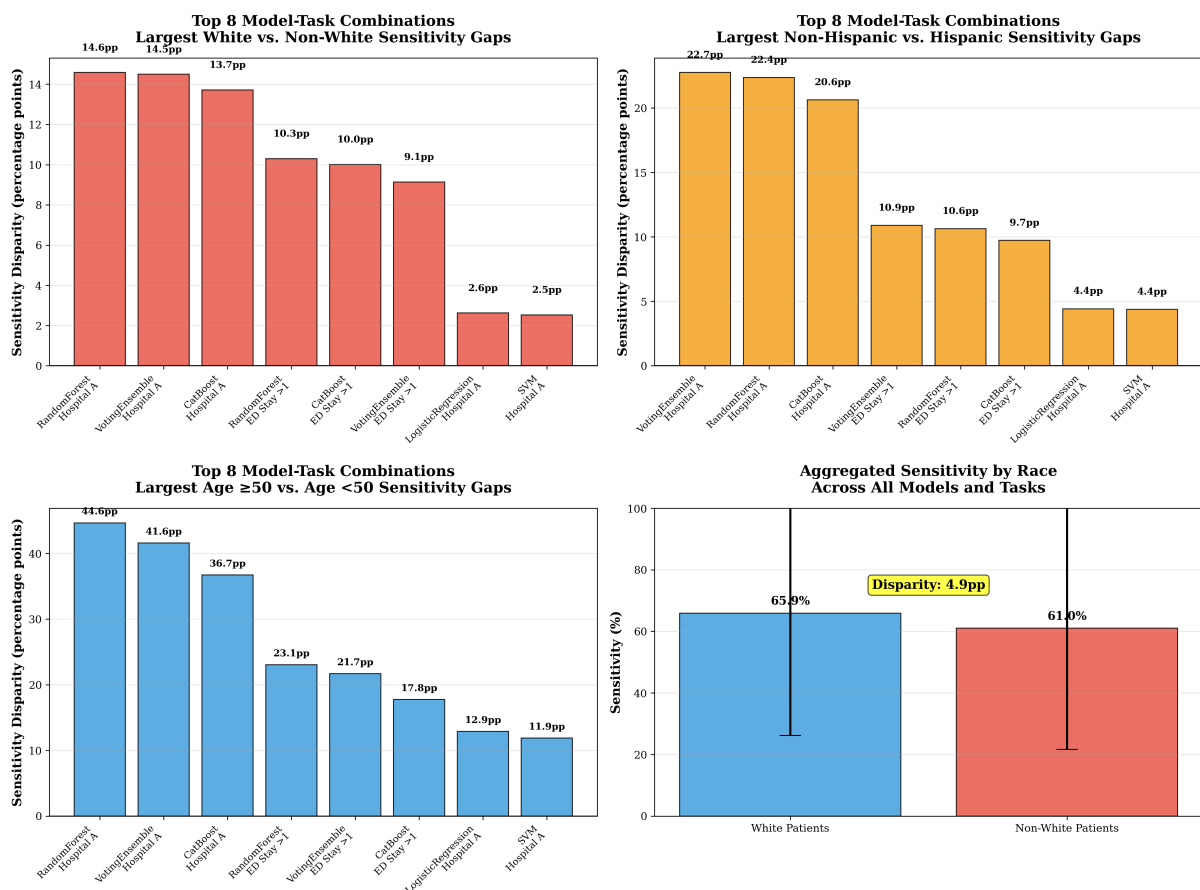


Figure S8: Comprehensive bias reporting visualizations showing sensitivity gaps across race/ethnicity, age, and insurance type for all prediction tasks.

4 CODE REPOSITORY DOCUMENTATION

All analysis code is available at: <https://www.github.com/Drglazizzo>

4.1 Software and Computational Environment

All analyses were conducted using Python 3.13 on a high-performance computing cluster. Key packages included:

- **Data Processing:** pandas 2.0, numpy 1.24, scikit-learn 1.3.2
- **Machine Learning:** xgboost 1.7.5, lightgbm 3.3.5, catboost 1.2.5, pytorch 2.0
- **Interpretability:** shap 0.43, lime 0.2
- **Visualization:** matplotlib 3.7, seaborn 0.12, plotly 5.15
- **Statistical Analysis:** scipy 1.11, statsmodels 0.14

4.2 Reproducibility Checklist

Table S13. Reproducibility checklist

Item	Status
MC-MED dataset access documented	Yes (PhysioNet Credentialing)
CITI training certification	Yes (Record ID: 72472937)
Random seeds fixed	Yes (seed = 42)
Software versions documented	Yes
Hyperparameters saved	Yes (Tables S10a, S10b, S10c)
Test set held out	Yes (11% chronological holdout)
Code version controlled	Yes (Git repository)